



浙江大学 信息与电子工程学院
College of Information Science & Electronic Engineering, Zhejiang University

How to Efficiently Balance Study and Contests at ISEE

ISEE 20th Grade Assembly Sharing

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Major : Information Engineering

Nov. 19, 2021

- All resource files have been uploaded to **BaiduNetdisk**

Link: https://pan.baidu.com/s/1tb8y5QGXwYeFC_o7D8zD9w Extraction code: ekn

- Scientific Research Training and Discipline Competition Management System

http://kyjs.zju.edu.cn/kyxl/login_tologin.action

- **ZJU Undergraduate Competition Recognition List (2021)**

http://kyjs.zju.edu.cn/kyxl/xtgg_getOne.action?ggid=931

- **Management Measures of Undergraduate Academic Competition of ZJU**

http://kyjs.zju.edu.cn/kyxl/xtgg_getOne.action?ggid=921

- **2021 ISEE Exempt Graduate Student Policy Document**

<http://www.isee.zju.edu.cn/2021/0724/c51195a2412141/page.psp>

- Annual selection of undergraduate competition special students in early September



Contests

[1] Zhejiang University Student Physics Innovation Competition (Provincial)

Not difficult, half for Physics(A) I and half for Physics(A) II. In 2019, it was first prize with 125 or more points, second prize with 95-125 points, and third prize with 80-95 points.

[2] Mathematics Competition for Colleges Students (preliminary provincial, final national)

Some of the first prize winners of the preliminaries can enter the final round (the winning rate of the preliminary round is as high as 70%). 1. Non-math group: calculus 2.

Mathematics group: mathematical analysis, advanced algebra, analytical geometry

[3] "ISEE-Rexchip" Mobile Innovation Competition (Embedded Practice)

2021 : <http://www.isee.zju.edu.cn/2021/0406/c51217a2276180/page.psp>

[4] RoboMaster MechaMaster Tournament (National)

Cross-combination practice in electronic control, vision, and hardware. ZJU's "Hello

World" team was the runner-up in the Super Match (Central Region), first prize in

Infantry Racing and Intelligent Shooting, first prize in Engineering Mining, and advanced to the national final in 2021.

[5] National Student Mathematical Modeling Competition (National/Provincial)

Rigorous and accurate thinking, some questions have standard reference answers.

(Past Outstanding Papers)

http://www.mcm.edu.cn/html_cn/block/8579f5fce999cdc896f78bca5d4f8237.html

(Past Outstanding Papers)

http://dxs.moe.gov.cn/zx/qkt/sxjm/qkt_sxjm_qgdxssxjmjsdh.shtml

Suggestion: learn from the past excellent papers in imitation and Google the basic methods of mathematical modeling while learning.

[6] MCM/ICM (National)

Extremely cost effective. High probability of winning.

Key points: good English expression of summary, clear and innovative modeling logic, good-looking diagrams.

Preparations: excellent English skills (Google/Baidu Translate alone ×), mastery of the basic numerical model.

[7] Electronic Design Competition for College Students (National)

**Happy = persistence in doing one's best + open-mindedness in listening to God's will.
(Similar to Embedded Artificial Intelligence Design Competition, Chip and System Design Competition)**

7.1 Courses: (1) review of the basics of digital and analog circuits (2) simulation, use of PCB software (3) precautions for welding (4) design and implementation of a simple system

7.2 Primary Period: (1) high-performance chip familiarization and practice (2) PCB manual board making (3) training of systems and modules

7.3 Self-Directed Training: (1) preparation of common modules (2) past tests training (3) resolving legacy issues (4) follow the forum(e.g., CC98) and major websites

[7] Electronic Design Competition for College Students (National)

7.4 Pipeline:

- (1) The list of devices will be published five or six days before the start of the national competition, so refer to the topics and lists of previous years to guess the modules that may be used and make targeted preparations.**
- (2) The group must discuss a complete and feasible plan before starting to divide the work, it is not wise to change the plan in the middle.**
- (3) You can use the Multisim simulation as a reference when calculating the parameters of the circuit, but do not rely on it completely, because in many cases the simulation and the actual will be very different.**
- (4) Set aside the last daylight hours to cascade the whole circuit for debugging, because often each sub-circuit works correctly independently, but cascading will have an impact.**

[1] Read MIT's “[How_To_Do_Research.pdf](#)” before doing research

[2] Learn to **WRITE** with the following references (third strong recommend)

Writing Style



Academic Phrasebank.pdf



Key Issues in Academic Writing.pdf



The-elements-of-style.pdf

[3] Learn to use LaTeX software(Overleaf/TexStudio), share my ZJUThesis templates

浙江大学

本科生课程论文



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递交日期 2021.11.20

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付潇 计算机视觉 2020-2021 冬 (宋明黎)

1 实验目的和要求

1.1 第一小节

以多元 Rosenbrock 函数为测试目标，编写基于 Wolfe 条件的线搜索算法，搜索方向包括最速下降和 Newton 法。
 $O(n^3)$

2 实验内容和原理

(简述实验有关的基本原理)

3 实验步骤与分析

(每个步骤结合对应部分的源代码分析)

4 实验环境及运行方法

(说明程序的编译环境和具体测试方法)

5 实验结果展示

(展示实验中的输入输出图像等)

6 心得体会

- Zhilin Yang : 28 years old young scientist, the methodology of an open life

<https://www.163.com/dy/article/FUSNC0S70511DPVD.html>

- [1] **We should prefer to do a simple idea.** The simpler the idea, the more essential.
- [2] When we do research, we should define **a high-level or long-term research agenda** for you, and then dismantle the problem.
- [3] Always realize that **the best things are yet to be made.**
- [4] **Be fast and iterative.**

"We do research, not every idea is correct, our Idea will always be wrong, and most people's most Idea is not Work. I have a rule that I write all my results in Google Spreadsheet, and then I find that **every time I write 400 or 500 lines or 1000 lines, I get a positive result.** So this means that the speed of the output depends on the speed of your iteration, and you have to iterate fast enough to get the results quickly. So I think that's a very important lesson."

- Hone basic skills such as PPT production and image processing (Ps, AI)
- From the organization clubs such as KAB, ASES, etc., I started to contact with the activities and seniors related to innovation and entrepreneurship
- Bravely participate in the college group's Internet+, Challenge Cup and other team recruiting, and don't feel inferior because you're not good at it
- Read books on economics (e.g, "Principles of Economics" and "Huawei Managing Change")
- Pay more attention to Hangzhou's startups and communicate more with others to feel the trend of the times.

- 2021年信息与电子工程学院创赛申报资料 (“青年红色筑梦之旅”赛道)
 - 护粟—基于5G物联网技术的智慧粮仓守护者
 - 晶鑫科技—中国光伏科技电池引领者
 - 浙江大学第十三届“蒲公英”大学生创业大赛项目汇总表.xlsx
- 2021年信息与电子工程学院创赛申报资料 (高教主赛道)
 - GoPrint++—多功能智能打印机先行者
 - HiLight—智能光磁交互屏首创者
 - 希谱科技—微型光谱仪民用化划时代革新者
 - 数字鱼—助力水产养殖数字经济
 - 欣心三维—面向未来的三维技术
 - 汇能科技—物联网时代的无线能源供给方案
 - 浙江大学第十三届“蒲公英”大学生创业大赛项目汇总表.xlsx
 - 火眼科技—新一代智慧社区安保管理解决方案



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浙大信电学院团队在“互联网+”省赛中再创佳绩！

Three gold (including the overall championship), one silver and one bronze in nine events

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> 若干“互联网+”国赛青年红色筑梦之旅赛道金奖	Aug 3, 2021 at 2:18 PM	--	Folder
> 若干“互联网+”高教主赛道金奖	Aug 3, 2021 at 2:18 PM	--	Folder

[1] Characteristics of ZJU's Innovation Education - Bottom-up and Situation-based

- **Space support:** In addition to Yuan space, there are nearly 70,000 square meters of entrepreneurial practice space open to students, such as E-works, Internet+ Creative Space, Cookie Creative Space, Purple Cow Creative Space, and Management Institute Incubator
- **Enterprise support:** for example, in 2016~2017, the university had 3.5 billion in research funding, of which 30% of the standing research funding came from society and enterprises, and preliminary statistics show that 80% of this 1.2 billion or so funding came from enterprises.
- **ZJU department alumni support:** 41% of Hangzhou entrepreneurs are from Zhejiang University. The ZJU department is quietly forming a force in the entrepreneurial community, with implication and support, alternating the roles of mentor and companion.
- **Failure culture:** People who have failed in entrepreneurship grow very fast and are very beneficial to the development of society

[1] A few points of the spirit of a good entrepreneur :

- (1) Not exclusively scientific decisions (based on data and calculations), but also based on creative imagination and intuitive judgment.
- (2) It is not the choice of means given a constraint, but the change of the constraint itself. It is not the choice of means given a constraint, but the change of the constraint itself.
- (3) Set goals that go beyond profit.

[2] "The Power of Company" series

- The power of the company should be the power of capital.
- The history of the market economy is told through the eyes of the company
- Need to recognize the costs and dangers of corporate change
- The unexpected inspiration of democracy and democratic institutions



[3] Tweet : "An entrepreneur's in-depth review: success, in line with the trend"

<https://mp.weixin.qq.com/s/Pu1ydcv8QnLnI3nVN5pogQ>

Don't regret anything in life. If it's good, it's wonderful.

If it's bad, it's experience! — Prof. Feynman

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